

When Audiences Can Tune Out: An Experimental Design for Estimating Heterogeneous Treatment Effects by Self-Selection Propensity

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Abstract

The persuasive effects of political messages depend on whether audiences choose to engage with them. However, experiments testing political messages typically require participants to view content, mechanically preventing self-selection and producing estimates that may systematically differ from those realized under voluntary viewing. I introduce a new experimental design, Continuous Heterogeneity Estimation for Self-Selection (CHESS), that recovers how persuasive effects vary across individuals with different selection propensities. In contrast to binary choice designs such as PICA, CHESS recovers continuous heterogeneous treatment effects while avoiding sensitivity to the researcher’s choice of control condition. In a preregistered experiment studying gun-control ads, I find that an anti-gun-control ad reduces support across the selection spectrum, while a pro-gun-control ad increases issue priority only among those most willing to view ads. These results illustrate how CHESS uncovers persuasive patterns obscured by average treatment effects.

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Experiments now play a key role in guiding political messaging decisions (Hewitt et al., 2024). For example, during the 2024 US presidential race, the Democratic super PAC Future Forward tested over 4,000 Harris ads to inform its \$700 million advertising effort, with the chief data scientist explaining that “for any given ad, you take a thousand people, split them into treatment and control, and then compare outcomes” (Klein, 2025; Schleifer and Goldmacher, 2024). This standard experimental design recovers the average treatment effect of exposure to a political ad. However, in practice, exposure to advertising involves self-selection: many viewers choose to tune out when ads come on. From a campaign’s perspective, the relevant question is not “How persuasive is this ad to a random sample of the population?” but rather “How persuasive is this ad to people who are likely to sit through it?”

This example highlights a broader methodological problem: in many settings, people must opt into interventions. In the domain of political communication, people choose whether or not to engage with ideologically incongruent news, open the door for a canvasser, or attend a political rally. More generally, people opt into government benefit programs or medical treatments. Across these contexts, standard experiments estimate average treatment effects, but researchers and practitioners may care about heterogeneity by the propensity to self-select into treatment: who is likely to opt into an intervention, and how effective is the intervention for those people?

I introduce a new experimental design—Continuous Heterogeneity Estimation for Self-Selection (CHESS)—that allows researchers to answer these kinds of questions, and I apply this design to estimate the efficacy of gun-control issue advertising. The main idea behind CHESS is to directly elicit a measure of participants’ willingness to self-select into treatment using the Becker–DeGroot–Marschak (BDM) method (Becker et al., 1964). Intuitively, this procedure asks “What is the minimum amount you would need to be paid in order to agree to treatment?” in such a way that participants are incentivized to answer truthfully. The dollar-valued answer provides a continuous measure of each participant’s propensity to self-select into treatment. Integrating this procedure into an experimental framework allows researchers to estimate how effects vary across the full selection distribution.

CHESS builds on a long-standing literature on self-selection in political science and medicine which proposes designs like the preference-incorporating choice and assignment (PICA) design and patient-preference trials (PPT) (Torgerson and Sibbald, 1998; de Benedictis-Kessner et al., 2019). These designs ask participants to choose between treatment and a control condition, allowing researchers to recover treatment effects for those who do and do not select

into treatment. While natural, these designs have two key drawbacks. First, choice-based designs recover one treatment effect per experimental arm, typically yielding just two heterogeneous effects in the binary case. Second, I show that the conclusions of choice-based designs can shift sharply, sometimes even reversing, depending on the desirability of the researcher-supplied control condition. CHESS overcomes these limitations by eliciting a continuous measure of the propensity to self-select into treatment prior to the experiment.

I illustrate CHESS with a preregistered online experiment studying pro- and anti-gun-control ads.¹ While the average treatment effect suggests that a pro-gun-control ad has little effect overall, CHESS shows that it increases the issue priority of gun control among the respondents most likely to view it. In contrast, an anti-gun-control ad consistently decreases support for gun control across the selection spectrum. These results show how CHESS can provide insights that conventional designs miss by enabling a more granular analysis of heterogeneity by selection propensity.

Self-Selection in Political Communication

Political communication channels typically involve a degree of selective exposure. People may choose whether to browse BlueSky or Truth Social, whether to click on an article after seeing a headline, or whether to listen to a candidate’s appearance on a podcast. Even in contexts where exposure is ostensibly unavoidable, such as with unskippable ads, people retain the choice of whether to pay attention or tune out.

Preferences for selective exposure to political information are well documented: people choose to consume news from ideologically congruent sources across many contexts (Taber and Lodge, 2006; Stroud, 2008; Kim, 2009; Iyengar and Hahn, 2009; Lavigne et al., 2025). It has long been theorized that preferences for selective exposure to news may shape voting decisions (Lazarsfeld et al., 1968). More recently, the rise of algorithmic social media feeds has raised concerns that preferences for selective exposure to ideologically congruent news may create ideological echo chambers contributing to political polarization (Iyengar et al., 2019; Ross Arguedas et al., 2022).

Despite the widespread acknowledgment that selective exposure mediates political communication, studies of the causal effect of communication interventions typically focus on average

¹This experiment was reviewed and approved by the University of Michigan IRB Health Sciences and Behavioral Sciences ID: HUM00275150. Preregistration is available at <https://aspredicted.org/fknm-9qxz.pdf>.

treatment effects using forced exposure designs ([Feldman \(2011\)](#); [Levendusky \(2013\)](#); [Broockman and Kalla \(2016\)](#); [Kalla and Broockman \(2022\)](#); [Broockman and Kalla \(2025\)](#)), but see [Kim and Patterson Jr \(2025\)](#)).

Yet, estimating how persuasion varies with self-selection propensity is both practically and theoretically important. Practically, understanding self-selection is valuable for determining how effective an intervention will be. For example, when considering whether to run an ad campaign, it is useful to know both how likely people are to watch an ad and how persuasive that ad would be to those who watch it.

Theoretically, understanding self-selection can help us disentangle correlation from causation when answering core questions about political communication and persuasion. As one illustration, [de Benedictis-Kessner et al. \(2019\)](#) argue that estimating heterogeneity by self-selection is crucial for disentangling whether the observed tendency to self-select into congenial media is a cause of polarization, or merely a reflection of it.

Existing Designs for Studying Self-Selection

A small but growing experimental design literature has developed methods for estimating heterogeneous treatment effects by the propensity to self-select into treatment.² I call this set of experimental designs choice designs, because the main idea behind them is to incorporate participant choice over treatment and control directly into the experiment. In a typical choice design, participants are first randomized into forced-exposure and selective-exposure arms. Participants in the forced-exposure arm are then randomly assigned to treatment or control, as in a standard experiment. Participants in the selective-exposure arm choose whether they would prefer to be assigned to treatment or control. Together, these arms allow researchers to estimate heterogeneous treatment effects for those who do and do not self-select into treatment.

The idea of incorporating choice via forced- and selective-exposure arms was first proposed by [Rücker \(1989\)](#), and has seen take-up in the medical literature, where it is referred to as a patient-preference trial ([Torgerson and Sibbald, 1998](#); [King et al., 2005](#)). A choice design was first applied to political communication by [Gaines and Kuklinski \(2011\)](#) who study attack ads. [Arceneaux et al. \(2012\)](#), [Arceneaux and Johnson \(2013\)](#), and [Levendusky](#)

²The broader problem of estimating heterogeneous treatment effects in the context of selection into treatment has long been formally studied in the econometrics literature ([Heckman and Smith, 1995](#); [Angrist et al., 1996](#)).

(2013) incorporate stated preferences into choice designs to study the effects of partisan media on attitudes toward the media, policy opinions and polarization, respectively. A more recent methodological improvement in this literature comes from [Knox et al. \(2019\)](#) and [de Benedictis-Kessner et al. \(2019\)](#) who formalize the incorporation of stated preferences into choice designs within a unified experimental framework called the Preference-Incorporating Choice and Assignment (PICA) design, an approach which enables sensitivity analysis with respect to measurement error.³

Directly incorporating participant choice into an experimental design is an intuitive and empirically successful way to estimate heterogeneity by self-selection propensity. However, choice-based experimental designs have two important drawbacks. First, they treat self-selection propensity as discrete rather than continuous. For example, in a PICA design studying those who prefer to consume entertainment or news, participants are classified into two discrete bins: those who prefer entertainment and those who prefer news (e.g., [Kim and Patterson Jr \(2025\)](#)). However, in reality, this kind of self-selection preference is likely to be continuous: some people may strongly prefer to watch entertainment content (or news), some may mildly prefer it, and some may be indifferent. By inferring a preference for entertainment or news content via a single choice, this kind of design forgoes collecting information on the intensity of the preference. Because choice designs treat self-selection propensity as discrete, they cannot detect nonlinearities in treatment effects as a function of self-selection propensity. I discuss one case where such treatment effects may arise in [Appendix C](#). In contrast, CHESS allows researchers to recover how treatment effects vary continuously with self-selection propensity.

A second issue with choice-based designs is that they are sensitive to the researcher-supplied control condition. For example, when testing the persuasiveness of a Fox News article, the researcher has a variety of potential control conditions to offer participants: they could choose a pure control task (e.g., do nothing for a fixed period of time), a moderate alternative (e.g., a CNN article), a democratic alternative (e.g., an MSNBC article), or even a fun alternative (e.g., an Onion article). While this may seem innocuous, it interacts in important ways with the choice design: by offering more attractive alternatives, the researcher can entice a larger share of participants to self-select out of reading the Fox News article. For some treatment-effect heterogeneity curves, the desirability of the outside option can change the

³In a concurrent working paper, [Levy et al. \(2025\)](#) deploys a similar BDM elicitation-based design to study persuasion in the context of the Israel-Gaza war. The two papers were developed independently and are complementary: this paper proposes a general methodological framework and connects it to previous choice-based designs, while [Levy et al. \(2025\)](#) provides additional evidence of the approach’s value in a distinct substantive domain.

sign and magnitude of the estimated heterogeneous treatment effects, thereby reversing the qualitative conclusions drawn from the choice experiment. I provide a hypothetical example of such a treatment effect reversal in [Figure 8](#), and I discuss this figure as well as the broader problem in more detail in [Appendix C](#). By estimating a measure of self-selection propensity that does not depend on the control condition, CHESS avoids this problem.

The CHESS Design

The CHESS design proceeds in three steps. First, participants are asked “What is the minimum amount of money you would need to be paid in order for you to participate in treatment?” in such a way that they are incentivized to answer truthfully. Second, most participants take part in a standard experiment. Finally, heterogeneous treatment effects are estimated using the minimum payment as a continuous proxy for self-selection propensity.

The key insight behind CHESS is that if each participant’s self-selection propensity were observable, then estimating heterogeneous treatment effects would be straightforward: after randomizing participants into treatment and control, we could estimate heterogeneous treatment effects just like we would for any other continuous observable (e.g., income, years of education).

The first step of CHESS is to elicit a proxy for self-selection propensity. CHESS reframes the question from “How likely would you be to select into treatment?” to the related question “What is the minimum amount you would need to be paid to participate in treatment?” The benefit of this reframing is that it allows the researcher to implement the question with real-stakes. Just like with choice designs, making participants answer a real-stakes question mitigates biases that arise from answering hypothetical questions ([Gaines and Kuklinski, 2011](#)). However, unlike choice designs, this question can be asked to every participant (not just those in a selective-exposure arm), and the answer is a continuous measure of self-selection propensity rather than a discrete one.

Directly asking participants how much they would need to be paid could cause mismeasurement, since participants might strategically under- or over-report their answer.⁴ To overcome this problem, CHESS elicits the minimum payment using the Becker–DeGroot–Marschak (BDM) procedure ([Becker et al., 1964](#); [Myerson, 1979](#)). In this procedure, a participant’s

⁴For example, participants may believe that overstating their minimum required payment would increase payment, or that understating it could increase the chance of selection.

reported minimum payment is compared to a randomly drawn wage. If the participant’s minimum payment is below the randomly drawn wage, then the participant is randomized into the experiment and paid the randomly drawn wage. The benefit of this procedure is that participants maximize their expected payoff by truthfully reporting the minimum amount that they would need to be paid. However, the cost is that the BDM mechanism is more difficult for participants to understand, so researchers should implement comprehension checks (Jack et al., 2022).⁵

The second step of the CHES design is to conduct an experiment. However, in order to avoid deception, participants who say that they need to be paid more than the randomly drawn wage must be excluded from the experiment.⁶ This has some benefits: it prevents the coercion of research subjects, and may remove subjects who would not comply with treatment from the experiment. However, it means that CHES cannot estimate treatment effects for participants with a very high aversion to treatment. This limitation is shared with choice designs, where participants who strongly prefer to avoid treatment may attrit when assigned to the forced-exposure arm.⁷

The third step of CHES is to compute heterogeneous treatment effects as normal. For example, researchers can discretize the minimum payment variable into K evenly sized bins to compute K heterogeneous treatment effects. Alternatively, the treatment effect can be estimated as a continuous function of the minimum payment.

Because CHES measures self-selection propensity without dividing the sample into forced- and selective-exposure arms, it can estimate these heterogeneous effects more efficiently than choice designs. In [Appendix A](#), I develop a potential outcomes framework that enables me to compare CHES and PICA (Neyman, 1923; Rubin, 1974). Using this framework, in [Appendix B](#), I show that CHES achieves the same precision for K heterogeneous treatment

⁵There are a few common willingness-to-pay elicitation procedures including standard BDM, multiple price lists, and binary search, each of which has costs and benefits. The stability and accuracy of willingness-to-pay elicitation mechanisms is a subject of discussion (Jack et al., 2022).

⁶As a practical note, the BDM procedure requires offering participants a price drawn from a random distribution. However, the vast majority of the weight of this distribution can be on the single price that the research team plans to pay participants. To avoid deception, a small number of participants can be offered another price, and ideally, the minimum and the maximum price described to participants should be offered.

⁷This variation of the CHES procedure assumes that participants would also participate in the control condition for the same price. An alternative approach would be to elicit minimum prices for participating in treatment and participating in control. Then, participants who would be willing to participate in both treatment and control could be randomized into the RCT. A benefit of this procedure is that the difference in price would provide a perhaps better measure of self-selection propensity. The main downsides to this measure are that it’s more complicated, and that it may induce researcher-demand biases since it necessarily discloses how the treatment differs from the control condition.

effects with $\frac{K}{6}$ times as much data under some simplifying assumptions. So, if both methods are used to compute heterogeneous treatment effects for participants with high and low self-selection propensity, then CHESS requires a third as much data.⁸

Evaluating Gun-Control Issue Ads with CHESS

Issue ads account for a meaningful fraction of political advertising dollars: [Falk et al. \(2006\)](#) find that \$404 million was spent on issue ads during the 108th congress. Yet, the prior literature on the effectiveness of television ads has largely focused on candidate ads ([Kalla and Broockman, 2022](#)). [Hall and Reynolds \(2012\)](#) theorize that issue ads may operate by increasing the salience of an issue or by changing public attitudes. [Kalla and Broockman \(2022\)](#) study ads regarding immigration and LGBTQ inclusion, and find that these ads are remembered but largely unpersuasive. [Junk and Rasmussen \(2024\)](#) study an advertising campaign about privacy risk, and similarly find null effects on attitudes and issue salience.

In this paper, I use CHESS to study issue ads from two prominent campaigns related to gun control. Participants in the pro-gun-control treatment view [Back-To-School Essentials](#), an ad created by Sandy Hook Promise. Participants in the anti-gun-control treatment view [Freedom’s Safest Place — Real Empowerment](#), an ad created by the National Rifle Association. In a placebo condition, participants view a [PSA about texting while driving](#) produced by the news station WJBF.

I conducted a preregistered online survey experiment on Prolific in September 2025.⁹ A total of 883 respondents were recruited, restricted to U.S. residents aged 18-65.¹⁰ Respondents who incorrectly answered comprehension questions regarding the BDM procedure twice were screened out of the survey. Participants were iteratively asked “Would you be willing to watch a 1 minute long advertisement for [wage]?” using a binary search procedure to search over wages between 10 cents and 2 dollars following an implementation of BDM suggested by

⁸To maintain the same degree of precision, the sample size of CHESS must linearly increase with the number of heterogeneous treatment effect bins. This relationship can be relaxed by making a functional form assumption regarding the shape of heterogeneous treatment effects (e.g. a second degree polynomial, as in [Figure 5](#)). In this case, all data can be applied to recover the parametric estimates, further improving efficiency.

⁹Preregistration is available at <https://aspredicted.org/fknm-9qxz.pdf>. All analyses I report are preregistered, though I do not preregister the validation exercise in [Figure 1](#). I preregister the main treatment effects and the main 3-bin self-selection heterogeneity results as “primary” analyses and all other results as “secondary” analyses.

¹⁰I also place a few additional technical restrictions on Prolific recruitment: I restrict to Prolific workers with a 99%+ approval rate on previous studies who have not participated in any of this study’s pilots.

Berkouwer and Dean (2022). As a validation exercise, Figure 1 shows that the BDM-elicited minimum payment correlates positively with self-reported propensity to skip ads, providing some evidence that the measure captures real variation in ad avoidance.¹¹ Participants were recruited into the experiment if they agreed to watch an ad for less than the randomly chosen wage. In practice, nearly all participants were offered a \$1.00 wage, which resulted in 823 of the 883 respondents being recruited into the experiment, though 11 participants attrited before outcome collection.¹² Recruited participants were randomly shown one of the ads. Table 1 documents covariate balance across treatment arms. To incentivize viewing the ad, payment was contingent on correctly answering two attention check questions related to the content of the ad. Finally, participants answered questions related to issue priority and attitudes towards gun control, and demographic information was collected.¹³

The results showcase a way in which average treatment effects can obscure pragmatically important heterogeneity. With respect to average treatment effects, I preregister hypotheses that pro-gun-control ads will increase support for gun control, anti-gun-control ads will decrease support, and that both ads will increase the issue priority of gun control. Figure 2 reports mean outcomes by treatment arm, and Table 2 reports average treatment effects. Overall, while all point estimates are directionally consistent with these hypotheses, neither ad has a statistically significant treatment effect on issue priority. The estimated effect of the pro-gun-control ad on support for gun control is close to zero (0.16 units, 95% CI: [-0.08, 0.40]). The anti-gun-control ad reduces support for gun control by 0.37 units (95% CI: [-0.61, -0.13]). This is a modest, but meaningful treatment effect: it represents 27% of the baseline difference between liberal and conservative attitudes on gun control (liberal mean: 6.27 vs. conservative mean: 4.94), or about one quarter of a standard deviation ($SD = 1.37$). So, if we evaluate these ads based on average treatment effects, we could conclude that the pro-gun-control ad was ineffective, and that the anti-gun-control ad modestly reduced support

¹¹The correlation is positive but modest, which is expected: the BDM measure and self-reported ad skipping capture related but distinct constructs. More broadly, validating how well laboratory measures of self-selection predict real-world behavior is an open question for both CHESS and choice designs, since both rely on lab elicitation to proxy for field behavior.

¹²The randomization procedure is \$1.00 with 99% probability, and \$2.00 with 1% probability. I do this so that the claim that the price is randomly chosen and the claim that the bonus may be up to \$2.00 are truthful.

¹³Issue attitude is the respondent’s support for gun control, as measured by a 7-point Likert scale question “In your opinion, should it be harder or easier to get a gun in this country?” with higher values indicating greater support for gun control. Issue priority is defined by the rank of “gun laws” in response to the question “Please rank the following issues from 1 (most important) to 5 (least important) when it comes to deciding your vote. The ranking should reflect how much the issue matters to you, no matter what position you personally take on it.” with the options (a) The Environment/Climate Change, (b) Immigration Policy, (c) Gun Laws, (d) Abortion Policy, and (e) The Economy. For presentation in figures and tables, I reverse the scale, so that higher values indicate greater importance.

for gun control.

However, heterogeneity by self-selection propensity tells a more nuanced story: the pro- and anti-gun-control ads appear to operate through different channels of persuasion. In [Figure 3](#) and [Table 3](#), I report heterogeneous treatment effects splitting propensity-to-select-into-treatment into three bins (high, medium and low propensity). While the treatment effects on support for gun control are largely consistent across selection propensity, the pro-gun-control ad increases issue priority by 0.33 ranks (95% CI: [0.01, 0.66]) only for high propensity respondents. This is a modest treatment effect, representing 28% of a standard deviation in the rank variable ($SD = 1.19$). [Table 4](#) shows that these results are robust to dropping respondents who fail the ad-related attention checks. This finding exemplifies the usefulness of CHES: overall treatment effects suggested that the pro-gun-control ad was ineffective, but the ad turns out to have a meaningful effect on issue priority for the respondents who are most likely to watch it according to this measure of self-selection propensity.

Measuring a proxy for self-selection propensity can also yield new insights when combined with other covariates. An analysis of heterogeneity by ideology alone in [Table 5](#) shows that the anti-gun-control ad is effective on conservatives, while the pro-gun-control ad is ineffective on both liberals and conservatives. [Figure 6](#) and [Table 6](#) interact political ideology with self-selection propensity. The interaction shows a large, albeit marginally significant, effect of the pro-gun-control ad on issue priority of 0.45 ranks (95% CI: [-0.06, 0.96]) for high propensity liberals. This finding demonstrates that self-selection heterogeneity can uncover insights about message targeting that heterogeneity by political ideology alone would miss.

One plausible reason for why the pro- and anti-gun-control ads operate through different channels of persuasion is ceiling effects. At baseline, participants are already pro-gun control: the control group mean is 5.68 on a 7-point scale, closest to the response that it should be “harder to get a gun.” In contrast, the average issue priority of gun control is in the middle of the scale, ranked at 2.95 out of 5. This suggests a hypothesis, not tested in this paper, that when attitude lobbying is not possible due to widespread support, campaigns may shift to persuasion on issue priority.

I conduct a variety of secondary analyses. [Figure 4](#) and [Figure 5](#) demonstrate two alternative approaches to analyzing heterogeneity by self-selection. [Figure 4](#) is a coarse 2-bin analysis, comparable to the PICA framework, while [Figure 5](#) is a continuous approach interacting treatment with a quadratic in self-selection propensity. While this study is not well-powered for estimating fine-grained heterogeneity, both figures suggest qualitatively similar conclu-

sions to the main heterogeneity analysis.

I report heterogeneity by a variety of other observables. The anti-gun-control ad is more effective at changing attitudes of politically disengaged participants, and less effective on those with a post-graduate education (Tables 7 and 10). Black participants respond much more strongly to both the anti-gun-control (on issue attitude) and pro-gun-control ads (on issue priority). I find no meaningful heterogeneity by gender (Table 9).

Figure 7 presents a simple analysis of participants’ free-responses to the question “Did you find the ad persuasive? Why or why not?” Responses were coded as persuasive or not persuasive using GPT-4, with coding validated against human coding for a random subset.¹⁴ Interestingly, respondents perceive the pro-gun-control ad as much more persuasive than the anti-gun-control ad, though the causal treatment effects tell a more complicated story.

Taken together, these results suggest issue ads can operate both by changing attitudes and by changing priorities among voters, providing empirical support for some mechanisms described in Hall and Reynolds (2012)’s theory of outside lobbying. These findings contrast with the small prior experimental literature on issue ads (Kalla and Broockman, 2022; Junk and Rasmussen, 2024). However, I study different ads about different issues, measure outcomes in the lab rather than the field, and find effects that vary with self-selection propensity, so many channels could drive the different findings. Beyond the specific findings about issue ads, these results demonstrate how the CHES design enables a flexible analysis of heterogeneity by the propensity to self-select into treatment which can yield practical insights that would not be recovered by a standard experimental design.

Conclusion

I propose a new experimental design, Continuous Heterogeneity Estimation for Self-Selection (CHES). This research design connects an old literature on willingness-to-pay elicitation with more recent work on leveraging participant choice to uncover heterogeneity by self-selection propensity (Becker et al., 1964; de Benedictis-Kessner et al., 2019).

CHES offers two advantages over existing designs. First, while choice designs restrict researchers to estimating binary heterogeneous treatment effects, CHES treats self-selection propensity as a continuous variable. This enables estimation of rich patterns of heterogeneity

¹⁴I hand-code a random subset of 100 responses. GPT-4 disagrees with my coding in 5 cases. An additional 4 cases are ambiguous, where GPT-4 and I agree but a reasonable coder might not.

including non-linear treatment effect curves that cannot be uncovered by previous designs. Second, CHESS elicits a proxy for self-selection propensity for every participant prior to randomization. This makes CHESS more efficient, requiring a third of the data to produce comparable estimates to choice designs. Additionally, it reduces the sensitivity of conclusions to the desirability of the control condition, an issue that can reverse the qualitative conclusions of choice designs. However, these advantages come at the cost of increased experimental complexity: CHESS requires the inclusion of additional comprehension checks, and is a more complicated design to implement.

I use the CHESS design to study issue ads, and find that they operate both by changing attitudes and by changing priorities among voters. I show how CHESS can uncover patterns of heterogeneity that change conclusions about the efficacy of ads. While average treatment effects imply that the pro-gun-control ad is ineffective, CHESS reveals that it increases the issue priority of gun control among the respondents who are most likely to view it. This experiment demonstrates that CHESS can provide researchers with more granular insights into treatment effect heterogeneity in contexts where participant self-selection is an important concern. Moreover, CHESS provides practitioners with a framework for identifying which audiences are both reachable and persuadable. This can help practitioners predict which interventions will succeed when deployed at scale, improving the external validity of experimental findings.

I highlight two directions for future research. First, validating how well laboratory measures of self-selection predict real-world behavior remains an open empirical question. Both choice designs and CHESS assume that choices in the lab predict choices in the field, but direct evidence for this assumption is limited.

Second, there are many plausible ways to refine the CHESS design. For example, it is not obvious that money is the most externally valid metric for measuring self-selection propensity: the BDM procedure could be extended to capture non-monetary costs such as time (e.g., “How long would you wait to avoid watching an ad?”). More broadly, the BDM procedure could be replaced with other per-participant behavioral proxies for self-selection propensity. Because lab experiments inform some of the highest stakes political communications decisions, it would be valuable to make progress on improving the external validity of these experimental designs.

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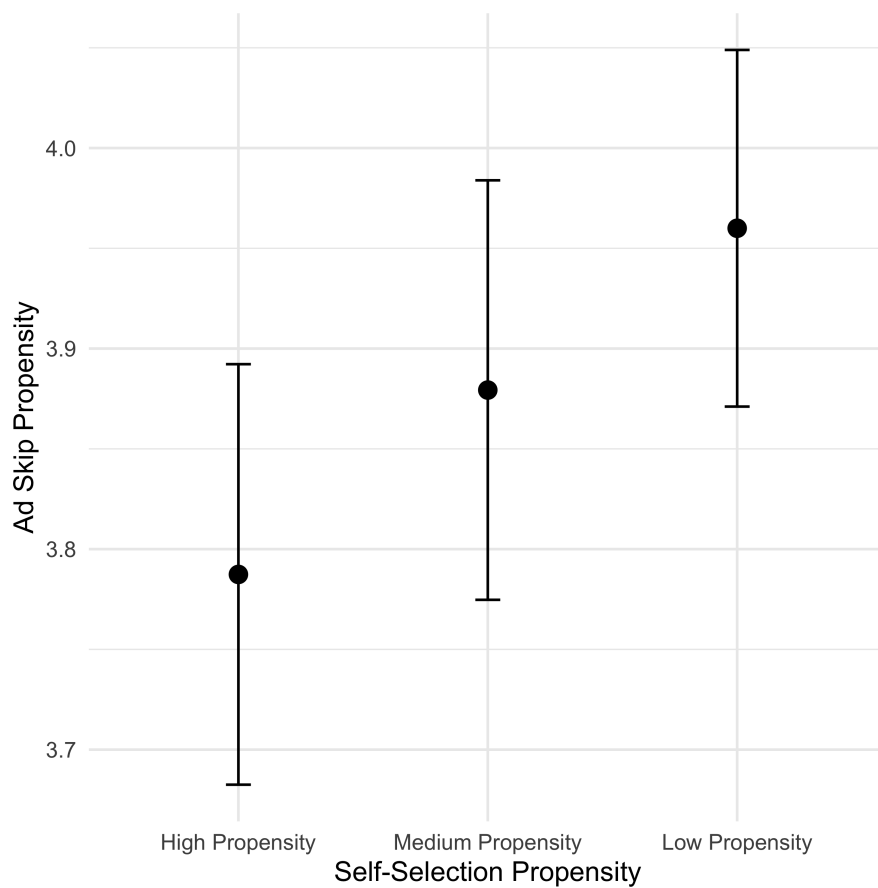
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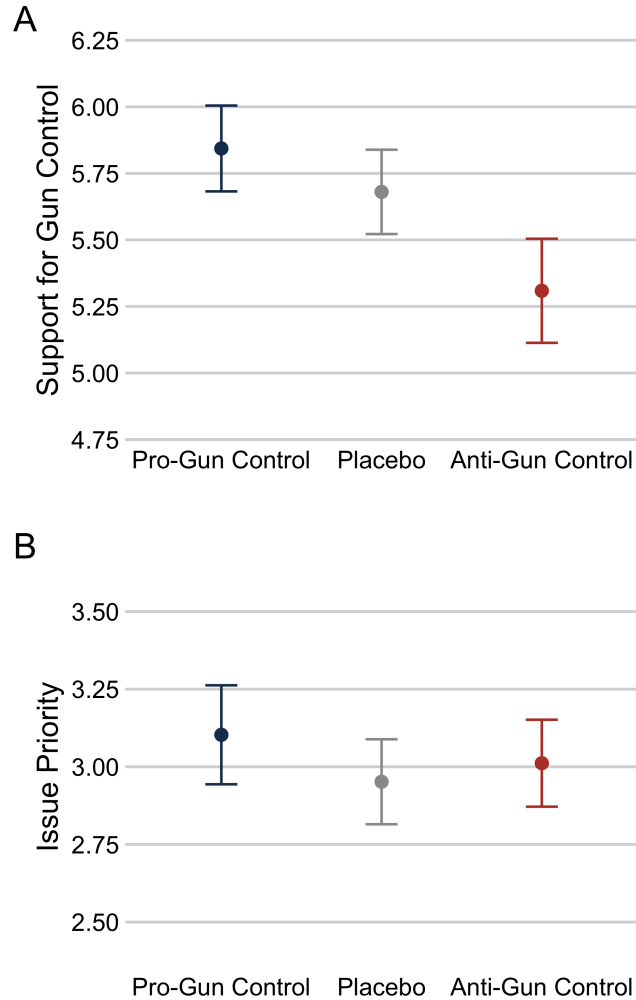
6 Figures

Figure 1: Self-Reported Ad-skip Propensity by Self-Selection Propensity



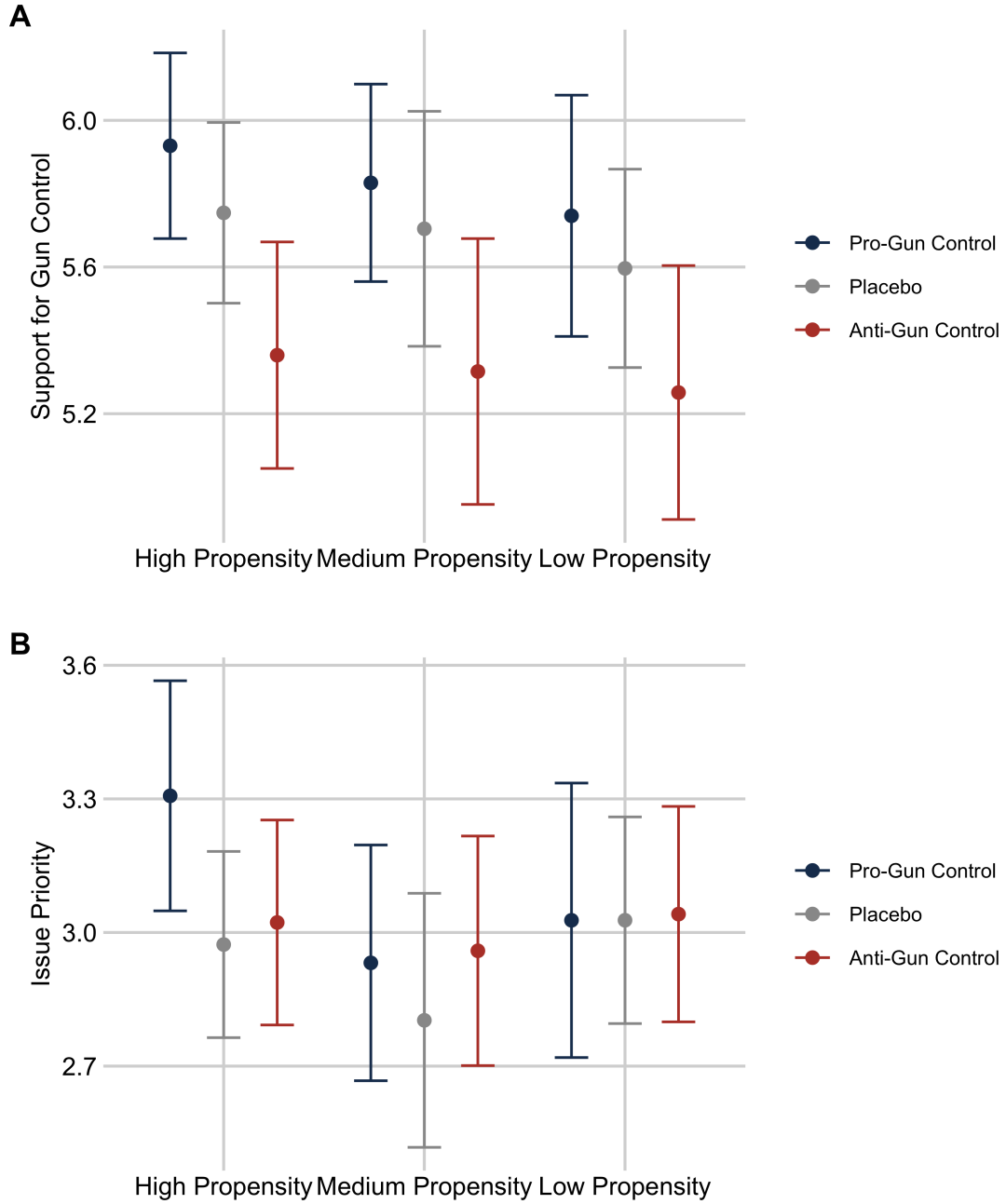
Notes: Self-reported ad skip propensity is measured on a Likert scale from “I never skip/ignore video ads” (1) to “I always skip/ignore video ads” (5). High, medium, and low propensity represent terciles of the participant’s elicited minimum payment to view an advertisement, where high propensity corresponds to the lowest payment. Points represent means, and error bars represent 95% confidence intervals.

Figure 2: Average Outcomes by Treatment Arm



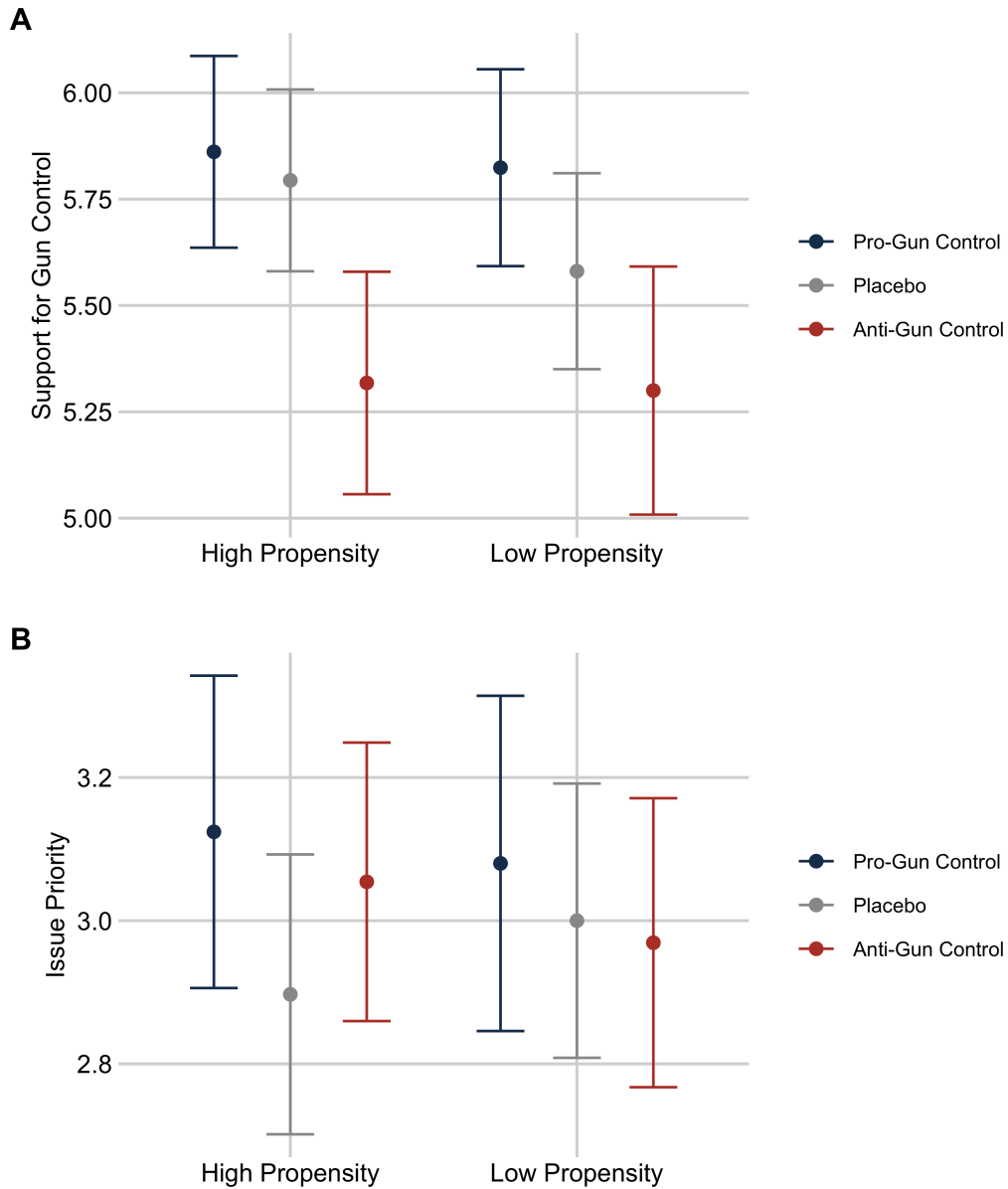
Notes: Panel A plots average support for gun control, which is the response to “In your opinion, should it be harder or easier to get a gun in this country?” on a 7-point Likert scale from “Much easier to get a gun” (1) to “Much harder to get a gun” (7). Panel B plots the average issue priority of gun control, which is the rank of “gun laws” in response to “Please rank the following issues from 1 (most important) to 5 (least important) when it comes to deciding your vote. The ranking should reflect how much the issue matters to you, no matter what position you personally take on it.” with the options (a) The Environment/Climate Change, (b) Immigration Policy, (c) Gun Laws, (d) Abortion Policy, and (e) The Economy. For graphing, the rank is reversed so that higher values represent more importance. Points represent means, and error bars represent 95% confidence intervals.

Figure 3: Heterogeneity by Self-Selection Propensity



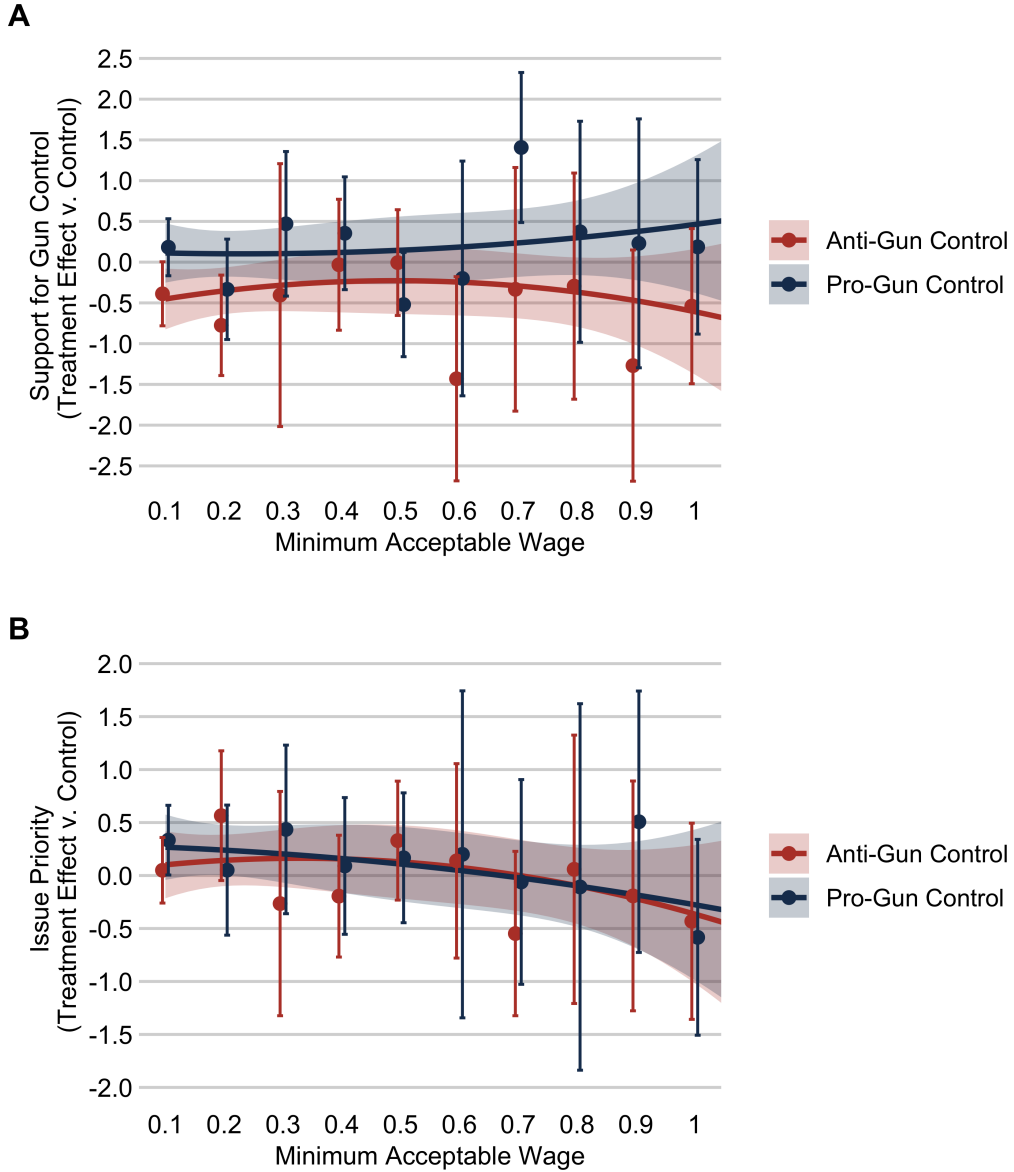
Notes: High, medium, and low propensity represent terciles of the participant's elicited minimum payment to view an advertisement, where high propensity corresponds to the lowest payment. Points represent means, and error bars represent 95% confidence intervals.

Figure 4: Heterogeneity by Self-Selection into Treatment (Binary)



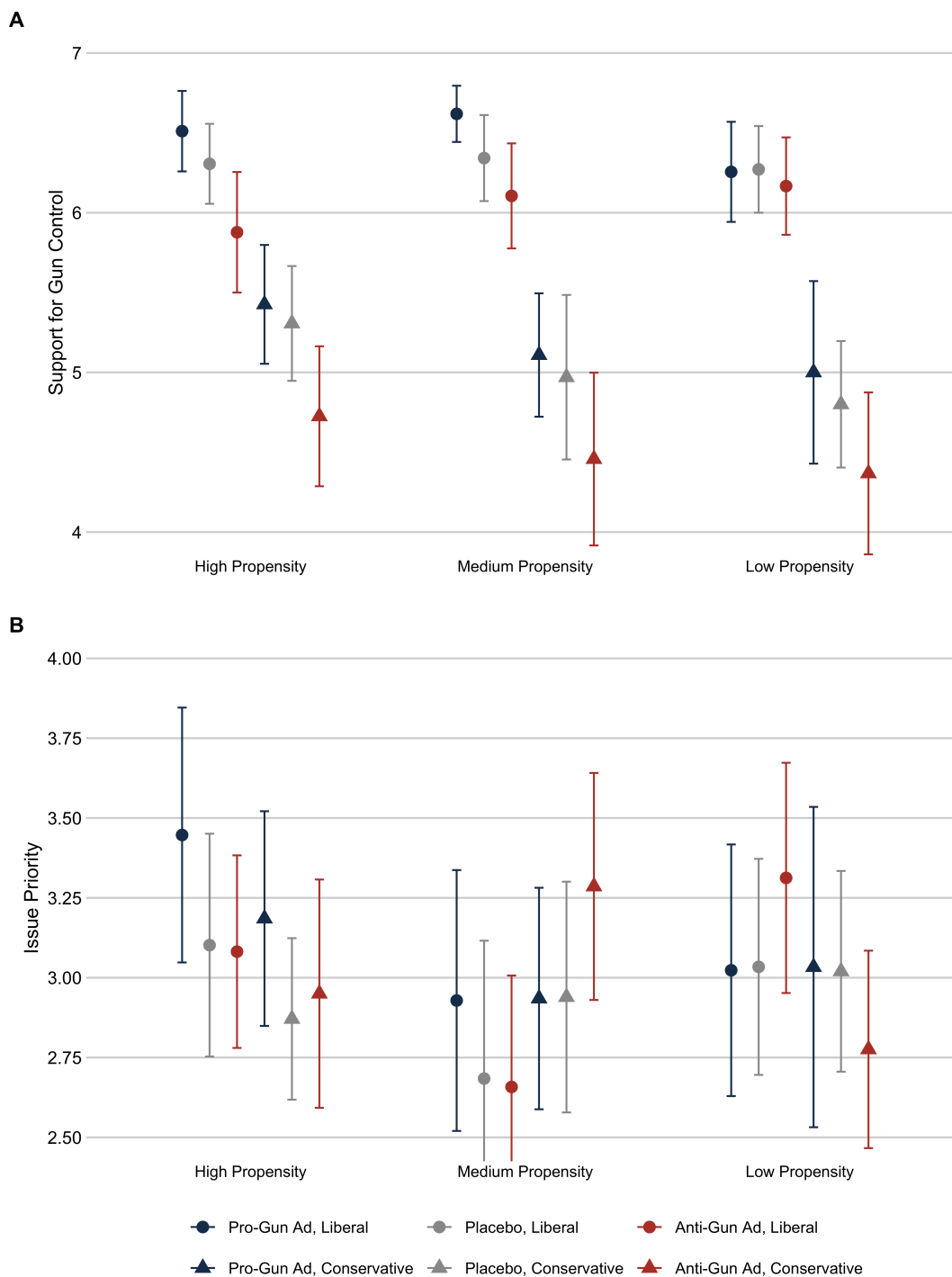
Notes: This figure replicates [Figure 3](#) with selection propensity divided into 2 bins rather than 3, making this approach more comparable to choice designs. Points represent means, and error bars represent 95% confidence intervals.

Figure 5: Heterogeneity by Self-Selection into Treatment (Parametric)



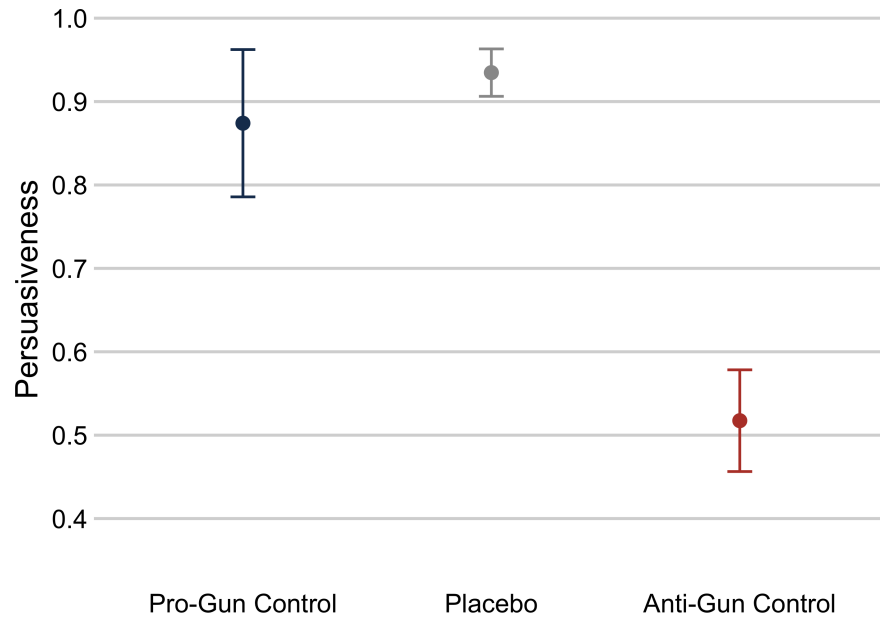
Notes: This figure plots treatment effects as a function of propensity to self-select into treatment. Curves are estimated using a second-degree polynomial interaction between treatment assignment and BDM elicited minimum acceptable wage. Points represent bin-level treatment effects, and error bars represent 95% confidence intervals.

Figure 6: Heterogeneity by Self-Selection and Political Ideology



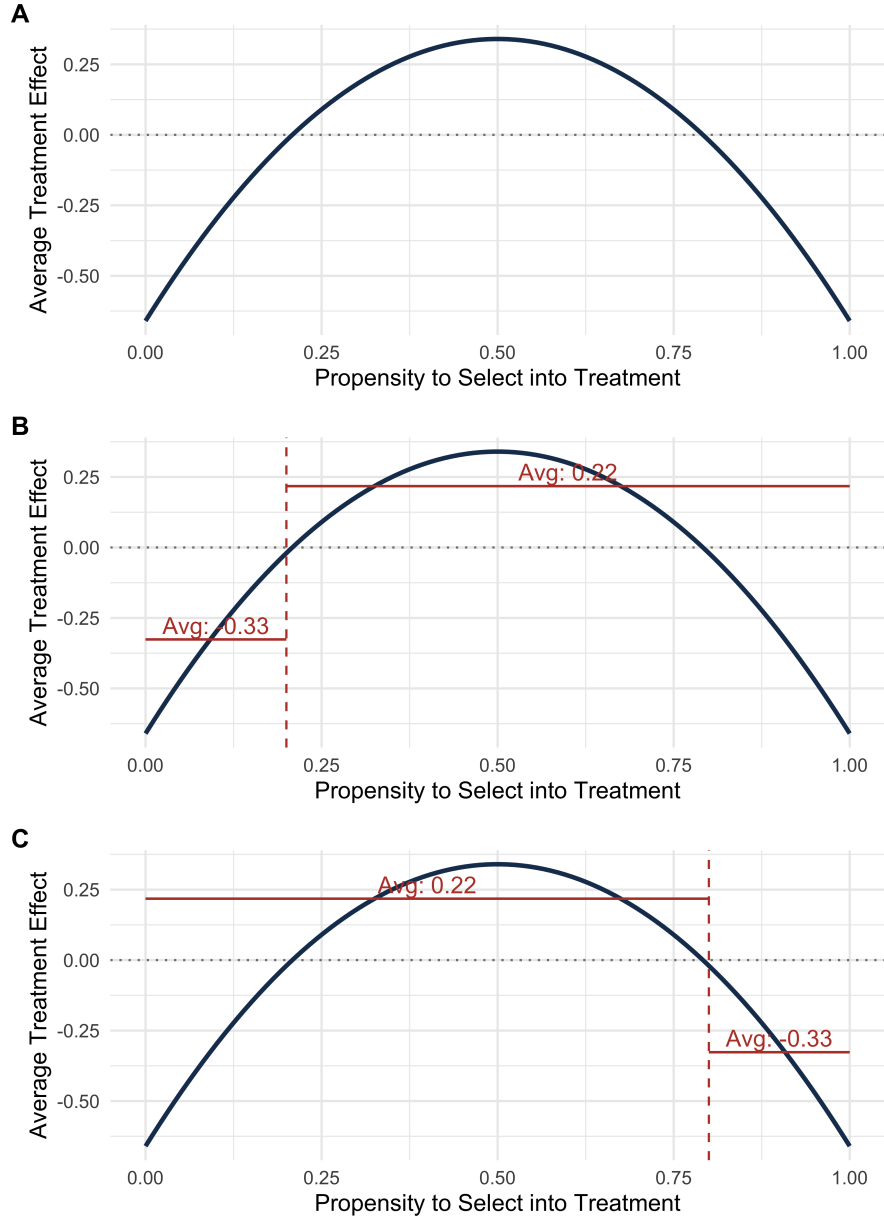
Notes: This figure plots mean outcomes by treatment arm, political ideology (liberal vs. conservative), and propensity to self-select into treatment in terciles. Points represent means, and error bars represent 95% confidence intervals.

Figure 7: Perceived Persuasiveness of Ads



Notes: This figure graphs participant-reported persuasiveness of the ad by treatment arm. Persuasiveness is coded from participant free responses to the question “Did you find the ad persuasive? Why or why not?” using GPT-4 as a classifier, where 1 represents persuasive and 0 represents not persuasive. Points represent means, and error bars represent 95% confidence intervals.

Figure 8: Hypothetical Heterogeneous Treatment Effects Curve



Notes: Panel A plots a hypothetical treatment effect curve as a function of propensity to self-select into treatment. Panels B and C illustrate how varying the outside option in a choice design produces different average treatment effect estimates across groups. Red dashed lines indicate the cutoff, and red horizontal lines represent average treatment effects on each side of the cutoff.

7 Tables

Table 1: Balance Table

	Control	Anti	Pro	Overall
<i>Demographics</i>				
Female	0.590 (0.029)	0.510 (0.031)	0.551 (0.030)	0.552 (0.017)
Black	0.113 (0.019)	0.106 (0.019)	0.150 (0.022)	0.123 (0.011)
Hispanic	0.078 (0.016)	0.080 (0.017)	0.086 (0.017)	0.081 (0.010)
Asian	0.126 (0.019)	0.106 (0.019)	0.120 (0.020)	0.118 (0.011)
Native	0.007 (0.005)	0.034 (0.011)	0.015 (0.007)	0.018 (0.005)
Prefer Not to Answer	0.007 (0.005)	0.008 (0.005)	0.019 (0.008)	0.011 (0.004)
<i>Education</i>				
High School	0.307 (0.027)	0.308 (0.029)	0.345 (0.029)	0.320 (0.016)
College	0.509 (0.029)	0.490 (0.031)	0.453 (0.031)	0.485 (0.017)
Post-Graduate	0.184 (0.023)	0.202 (0.025)	0.202 (0.025)	0.196 (0.014)
<i>Politics</i>				
Ideology	3.519 (0.104)	3.436 (0.115)	3.374 (0.107)	3.446 (0.063)
Political Engagement	4.220 (0.139)	4.220 (0.146)	4.252 (0.149)	4.230 (0.083)
Observations	293	263	267	823

Notes: Entries are sample means of pre-treatment covariates by treatment arm, with standard errors in parentheses. Ideology ranges from 1 (very liberal) to 7 (very conservative). Political engagement ranges from 0 (Never) to 7 (Every day of the week).

Table 2: Average Treatment Effects

Dependent Variables:	Attitude		Issue Priority	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Constant	5.680*** (0.0844)	6.493*** (0.1716)	2.952*** (0.0715)	2.837*** (0.1713)
Anti-Gun Control	-0.3715*** (0.1231)	-0.3554*** (0.1048)	0.0597 (0.1042)	0.0790 (0.1046)
Pro-Gun Control	0.1631 (0.1227)	0.1150 (0.1042)	0.1512 (0.1039)	0.1489 (0.1041)
Controls	False	True	False	True
<i>Fit statistics</i>				
R ²	0.02281	0.30991	0.00263	0.02145
Observations	812	812	812	812

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Issue attitude is participants' response on a 7-point Likert scale to the question "In your opinion, should it be harder or easier to get a gun in this country?" with higher values indicating greater support for gun control. Issue priority is participants' reported rank (1-5) of gun laws relative to the environment/climate change, immigration policy, abortion policy, and the economy, rescaled so higher values indicate higher priority. Controls: race, gender, educational attainment, political ideology, and degree of political engagement.

Table 3: Heterogeneous Treatment Effects by Self-Selection Propensity

Dependent Variable:		Attitude				
Selection Propensity		High		Medium		Low
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	5.748*** (0.1296)	6.256*** (0.2910)	5.704*** (0.1677)	6.774*** (0.3010)	5.596*** (0.1484)	6.679*** (0.3053)
Anti-Gun Control	-0.3882** (0.1943)	-0.4216** (0.1774)	-0.3892* (0.2355)	-0.3729* (0.1980)	-0.3386 (0.2163)	-0.2968* (0.1776)
Pro-Gun Control	0.1829 (0.1878)	0.1702 (0.1686)	0.1253 (0.2254)	0.1323 (0.1876)	0.1434 (0.2344)	-0.1174 (0.1928)
Controls	False	True	False	True	False	True
<i>Fit statistics</i>						
R ²	0.02786	0.25858	0.02379	0.36829	0.01607	0.37649
Observations	301	301	232	232	279	279

Dependent Variable:		Issue Priority				
Selection Propensity		High		Medium		Low
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	2.973*** (0.1130)	2.836*** (0.2858)	2.803*** (0.1437)	2.701*** (0.3097)	3.028*** (0.1205)	3.002*** (0.3037)
Anti-Gun Control	0.0495 (0.1694)	0.1077 (0.1743)	0.1561 (0.2018)	0.1754 (0.2037)	0.0137 (0.1756)	0.0507 (0.1767)
Pro-Gun Control	0.3340** (0.1637)	0.3830** (0.1656)	0.1290 (0.1931)	0.1708 (0.1931)	-0.0001 (0.1902)	-0.0681 (0.1918)
Controls	False	True	False	True	False	True
<i>Fit statistics</i>						
R ²	0.01550	0.04663	0.00301	0.07021	0.00003	0.04812
Observations	301	301	232	232	279	279

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Selection propensity is a tercile split of the minimum reported wage to watch a one minute ad, where high propensity corresponds to a low minimum wage. Controls: race, gender, educational attainment, political ideology, and degree of political engagement.

Table 4: HTEs by Selection: Passed Attention Check Subsample

Dependent Variable:		Attitude				
Selection Propensity		High		Medium		Low
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	5.767*** (0.1363)	6.224*** (0.3240)	5.725*** (0.1732)	7.022*** (0.3245)	5.634*** (0.1560)	6.558*** (0.3231)
Anti-Gun Control	-0.2817 (0.2162)	-0.4005** (0.1985)	-0.4787* (0.2529)	-0.4138* (0.2116)	-0.4798** (0.2363)	-0.4702** (0.1940)
Pro-Gun Control	0.1768 (0.2003)	0.2144 (0.1784)	0.0425 (0.2416)	0.0419 (0.2007)	0.0599 (0.2529)	-0.2054 (0.2070)
Controls	False	True	False	True	False	True
<i>Fit statistics</i>						
R ²	0.01626	0.26385	0.02532	0.38382	0.02251	0.38641
Observations	260	260	203	203	241	241

Dependent Variable:		Issue Priority				
Selection Propensity		High		Medium		Low
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	2.961*** (0.1176)	2.569*** (0.3136)	2.797*** (0.1433)	2.566*** (0.3219)	3.010*** (0.1293)	3.029*** (0.3287)
Anti-Gun Control	0.0829 (0.1865)	0.2118 (0.1922)	0.2685 (0.2093)	0.2974 (0.2099)	0.0157 (0.1959)	0.0774 (0.1974)
Pro-Gun Control	0.3085* (0.1728)	0.3768** (0.1727)	0.1344 (0.1999)	0.1850 (0.1991)	-0.0260 (0.2097)	-0.0596 (0.2106)
Controls	False	True	False	True	False	True
<i>Fit statistics</i>						
R ²	0.01274	0.06982	0.00817	0.09840	0.00015	0.05504
Observations	260	260	203	203	241	241

Notes: This table replicates [Table 3](#) on the subset of participants who pass both attention checks regarding the content of the ad that they watch.

Table 5: Heterogeneous Treatment Effects by Political Ideology

Dependent Variables:	Attitude		Issue Priority	
Political Ideology	Liberal	Conservative	Liberal	Conservative
Model:	(1)	(2)	(3)	(4)
Constant	5.718*** (0.1642)	4.551*** (0.2365)	2.895*** (0.2136)	2.831*** (0.1849)
Anti-Gun Control	-0.2583** (0.1185)	-0.4452** (0.1840)	0.0961 (0.1542)	0.0576 (0.1438)
Pro-Gun Control	0.1263 (0.1196)	0.1916 (0.1802)	0.1754 (0.1556)	0.1298 (0.1408)
Controls	True	True	True	True
<i>Fit statistics</i>				
R ²	0.08137	0.10489	0.02897	0.02977
Observations	413	399	413	399

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controls: race, gender, educational attainment, and degree of political engagement.

Table 6: Heterogeneous Treatment Effects by Self-Selection $\times$ Political Ideology

Dependent Variable:		Attitude				
Political Ideology		Liberal			Conservative	
Selection Propensity	High	Medium	Low	High	Medium	Low
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	5.507*** (0.3125)	6.046*** (0.2522)	5.777*** (0.2968)	4.724*** (0.3777)	4.704*** (0.4588)	4.464*** (0.4288)
Anti-Gun Control	-0.3959* (0.2178)	-0.2826 (0.1917)	-0.1628 (0.2112)	-0.4649 (0.3022)	-0.5004 (0.3912)	-0.2896 (0.3136)
Pro-Gun Control	0.1516 (0.2201)	0.2799 (0.1880)	-0.0631 (0.2167)	0.1900 (0.2689)	0.1186 (0.3696)	0.1664 (0.3597)
Controls	True	True	True	True	True	True
<i>Fit statistics</i>						
R ²	0.14808	0.14634	0.07868	0.11189	0.05367	0.20579
Observations	145	118	150	156	114	129

Dependent Variable:		Issue Priority				
Political Ideology		Liberal			Conservative	
Selection Propensity	High	Medium	Low	High	Medium	Low
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	2.788*** (0.3709)	2.917*** (0.3911)	2.820*** (0.3658)	2.521*** (0.3097)	2.880*** (0.3367)	3.164*** (0.3364)
Anti-Gun Control	0.0793 (0.2585)	0.0531 (0.2972)	0.2825 (0.2603)	0.1500 (0.2478)	0.2819 (0.2871)	-0.1773 (0.2460)
Pro-Gun Control	0.4508* (0.2613)	0.3002 (0.2915)	-0.1081 (0.2671)	0.3614 (0.2205)	-0.0180 (0.2712)	-0.0284 (0.2821)
Controls	True	True	True	True	True	True
<i>Fit statistics</i>						
R ²	0.08008	0.07527	0.08020	0.05161	0.09690	0.07912
Observations	145	118	150	156	114	129

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controls: race, gender, educational attainment, and political engagement.

Table 7: Heterogeneous Treatment Effects by Political Engagement

Dependent Variables:		Attitude			Issue Priority		
Political Engagement	0–3 Days	4–6 Days	7 Days	0–3 Days	4–6 Days	7 Days	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	
Constant	6.272*** (0.2650)	6.706*** (0.2637)	6.916*** (0.2365)	2.782*** (0.2465)	2.663*** (0.2861)	2.919*** (0.2407)	
Anti-Gun Control	-0.4456** (0.1738)	-0.4051** (0.1931)	-0.2493 (0.1826)	0.0764 (0.1617)	0.2382 (0.2094)	-0.0302 (0.1858)	
Pro-Gun Control	0.1094 (0.1749)	-0.0278 (0.1885)	0.2052 (0.1785)	0.1562 (0.1627)	0.1421 (0.2045)	0.1118 (0.1817)	
Controls	True	True	True	True	True	True	
<i>Fit statistics</i>							
R ²	0.24192	0.36564	0.38948	0.06051	0.04078	0.05312	
Observations	343	209	260	343	209	260	

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controls: race, gender, educational attainment, and political ideology.

Table 8: Heterogeneous Treatment Effects by Race

Dependent Variables:	Attitude			Issue Priority		
Race	White	Black	Other	White	Black	Other
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Constant	6.516*** (0.1957)	6.223*** (0.5110)	6.551*** (0.4245)	2.809*** (0.1921)	2.638*** (0.5465)	3.222*** (0.4260)
Anti-Gun Control	-0.2818** (0.1235)	-0.9027*** (0.3054)	-0.3718 (0.2552)	0.0666 (0.1213)	0.2228 (0.3265)	0.2379 (0.2562)
Pro-Gun Control	0.1911 (0.1246)	0.0287 (0.2790)	-0.0509 (0.2508)	-0.0525 (0.1223)	0.7347** (0.2983)	0.5555** (0.2517)
Controls	True	True	True	True	True	True
<i>Fit statistics</i>						
R ²	0.33001	0.23707	0.27080	0.00621	0.12836	0.07658
Observations	579	101	144	579	101	144

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controls: gender, educational attainment, political ideology, and degree of political engagement.

Table 9: Heterogeneous Treatment Effects by Gender

Dependent Variables:	Attitude		Issue Priority	
Gender	Female	Male	Female	Male
Model:	(1)	(2)	(3)	(4)
Constant	6.918*** (0.1953)	6.407*** (0.2658)	3.054*** (0.2093)	2.740*** (0.2508)
Anti-Gun Control	-0.2855** (0.1318)	-0.3832** (0.1684)	0.1024 (0.1412)	0.0700 (0.1589)
Pro-Gun Control	0.1085 (0.1293)	0.1854 (0.1708)	0.1172 (0.1385)	0.1648 (0.1611)
Controls	True	True	True	True
<i>Fit statistics</i>				
R ²	0.26786	0.35263	0.02727	0.01469
Observations	454	358	454	358

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controls: race, educational attainment, political ideology, and degree of political engagement.

Table 10: Heterogeneous Treatment Effects by Educational Attainment

Dependent Variables:		Attitude			Issue Priority		
Education	Post-Grad	College	HS/Below	Post-Grad	College	HS/Below	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	
Constant	6.392*** (0.3865)	6.973*** (0.2245)	6.147*** (0.3067)	3.052*** (0.4050)	2.943*** (0.2236)	2.550*** (0.3020)	
Anti-Gun Control	-0.0622 (0.2295)	-0.4074*** (0.1452)	-0.3831* (0.2015)	0.1382 (0.2405)	0.0656 (0.1446)	0.0783 (0.1984)	
Pro-Gun Control	0.3567 (0.2287)	-0.0643 (0.1470)	0.2543 (0.1953)	0.1608 (0.2396)	0.1982 (0.1464)	-0.0192 (0.1923)	
Controls	True	True	True	True	True	True	
<i>Fit statistics</i>							
R ²	0.35909	0.32185	0.31516	0.05141	0.03917	0.05285	
Observations	161	399	252	161	399	252	

Notes: OLS estimates. IID standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Controls: race, gender, political ideology, and degree of political engagement.

A Potential Outcomes Statistical Framework Connecting PICA and CHESS

In order to more precisely compare the CHESS and PICA designs, I now formalize the estimation problem using a potential outcomes framework (Neyman, 1923; Rubin, 1974). Let $X_i \in \{0, 1\}$ indicate whether individual i receives treatment, let Y_i denote individual i 's observed outcome, and let $C_i \in \{0, 1\}$ denote whether individual i chooses to participate in the intervention if given a choice. Let $Y_i(1)$ and $Y_i(0)$ denote individual i 's potential outcomes under treatment and control, respectively.

In a standard PICA design, participants are randomly assigned to one of two arms:

1. A *choice arm*, where participants self-select into the intervention (C_i is observed and determines X_i);
2. A *forced-assignment arm*, where participants are assigned to treatment or control regardless of preference (X_i is randomly assigned).

This design allows us to identify the following treatment effects:

1. $E[Y_i(1) - Y_i(0)]$, the average treatment effect
2. $E[Y_i(1) - Y_i(0) \mid C_i = 1]$, the treatment effect among participants who would opt into treatment if given a choice
3. $E[Y_i(1) - Y_i(0) \mid C_i = 0]$, the treatment effect among participants who would opt into control if given a choice

Let z denote a participant's willingness-to-select-into-treatment, which we can think of as their anticipated utility from participation.

In the choice arm, the researcher offers participants the choice to either opt into treatment or to select an outside option. For now, assume that all participants value the outside option at a fixed level, d . Then,

$$C_i = \mathbb{I}[z_i > d]$$

That is, participants opt into treatment if they like treatment more than the outside option. Implicitly, the provided alternative divides participants into high- and low-willingness-to-select-into-treatment groups. Then, the PICA heterogeneous treatment effects correspond

to

$$E[Y_i(1) - Y_i(0) \mid z > d]$$

$$E[Y_i(1) - Y_i(0) \mid z \leq d],$$

where the value of d depends on how desirable the researcher-supplied outside option is to participants.

Suppose that the elicitation procedure provides a valid measure of participants' willingness-to-select-into-treatment. That is, assume that the answer to "How much would I need to pay you to participate in treatment?" perfectly correlates with willingness-to-select-into-treatment z_i . Then, we can compute

$$E[Y_i(1) - Y_i(0) \mid z \in (a, b)]$$

for any interval (a, b) with empirical support in the data. In particular, we can recover the PICA estimates, as well as any other intervals of interest.

In practice, we may want to provide a non-parametric estimate for how the treatment effect varies with willingness-to-select-into-treatment. To do this, we can define cutpoints along z that divide the data into K evenly sized bins, and then can compute K within-bin treatment effects. In this way, CHESS allows researchers to estimate how treatment effects vary across the full distribution of selection propensity, rather than being restricted to a binary analysis as in PICA.

B Precision of CHESS/PICA Estimators

This section derives back-of-the-envelope standard errors for (i) a CHESS estimator that reports K heterogeneous treatment effects (HTEs) and (ii) a binary-treatment PICA estimator that reports two choice-specific HTEs (those who would select into treatment vs. those who would not). I'll show that, under some simplifying assumptions, CHESS requires $\frac{K}{6}$ times as many observations to achieve the same degree of precision as PICA for each HTE.

B.1 Standard Error of Treatment Effect Estimator

First, I will derive the standard error of a treatment effect estimator. Let $X \in \{0, 1\}$ indicate the treatment condition, and let Y be the outcome variable. Suppose we have N observations, randomized evenly into treatment and control. The treatment effect estimate is given by

$$\tau = \mathbb{E}[Y|X = 1] - \mathbb{E}[Y|X = 0]$$

with estimator

$$\hat{\tau} = \frac{2}{N} \sum_{i=1}^N y_i \mathbb{I}[X = 1] - \frac{2}{N} \sum_{i=1}^N y_i \mathbb{I}[X = 0].$$

Then,

$$\text{Var}(\hat{\tau}) = \text{Var}(\bar{Y}_1 - \bar{Y}_0) = \text{Var}(\bar{Y}_1) + \text{Var}(\bar{Y}_0).$$

Define

$$\sigma_0^2 = \text{Var}(Y|X = 0) \quad \sigma_1^2 = \text{Var}(Y|X = 1).$$

Then,

$$\text{Var}(\bar{Y}_1) = \frac{2\sigma_1^2}{N} \quad \text{Var}(\bar{Y}_0) = \frac{2\sigma_0^2}{N}$$

Assuming homoskedasticity of errors,

$$\sigma^2 = \sigma_0^2 = \sigma_1^2$$

so

$$\text{Var}(\hat{\tau}) = \text{Var}(\bar{Y}_1) + \text{Var}(\bar{Y}_0) = \frac{2\sigma^2}{N} + \frac{2\sigma^2}{N} = \frac{4\sigma^2}{N}$$

and

$$\text{SE}(\hat{\tau}) = 2\sigma\sqrt{\frac{1}{N}}.$$

B.2 Efficiency of CHESSE Estimator

Now, I will derive the standard error for heterogeneous treatment effects when we divide N observations into K equal sized bins, representing K equal cuts along some dimension of heterogeneity Z . In each bin, there are $\frac{N}{K}$ observations. As a simplifying assumption, I impose homoskedasticity of errors within each of these bins. Formally, $\forall k \in K$,

$$\sigma^2 = \text{Var}(Y \mid X = 0, Z \in \text{bin } k) = \text{Var}(Y \mid X = 1, Z \in \text{bin } k)$$

Then,

$$\text{Var}(\hat{\tau}_k) = \frac{4K\sigma^2}{N},$$

and

$$\text{SE}(\hat{\tau}_k) = 2\sigma\sqrt{\frac{K}{N}}.$$

So, if a researcher divides the willingness-to-select spectrum into K equally-sized bins and collects N_{CHESSE} observations, then

$$\text{SE}(\hat{\tau}_{\text{CHESSE}}) = 2\sigma\sqrt{\frac{K}{N_{\text{CHESSE}}}}.$$

B.3 Efficiency of PICA Estimator

Following [Knox et al. \(2019\)](#), let $C \in \{0, 1\}$ indicate the treatment condition that a participant would choose if given the option. Then, the estimator for the PICA heterogeneous treatment effects in the binary case is given by

$$\tau_1 = \mathbb{E}[Y(1) - Y(0) \mid C = 1] = \frac{\mu_{\text{sel}} - \mu_{\text{forced},0}}{p},$$

and

$$\tau_0 = \mathbb{E}[Y(1) - Y(0) \mid C = 0] = \frac{\mu_{\text{forced},1} - \mu_{\text{sel}}}{1 - p},$$

where p represents the probability that participants select into treatment in the selective-exposure arm, and C represents the choice that participants make in the selective-exposure

arm. This estimator can be found after equation four in [Knox et al. \(2019\)](#) and matches the estimator in [Gaines and Kuklinski \(2011\)](#).

Let total PICA sample size be N_{PICA} . Assume equal allocation across arms:

$$\text{forced arm size} = \frac{N_{\text{PICA}}}{2}, \quad \text{selective arm size} = \frac{N_{\text{PICA}}}{2}.$$

Within the forced arm, assume treatment is randomized 50/50, so

$$n_{1,\text{forced}} = \frac{N_{\text{PICA}}}{4}, \quad n_{0,\text{forced}} = \frac{N_{\text{PICA}}}{4}.$$

To get PICA estimators, replace the population means with sample means:

$$\hat{\tau}_1 = \frac{\hat{\mu}_{\text{sel}} - \hat{\mu}_{\text{forced},0}}{p}, \quad \hat{\tau}_0 = \frac{\hat{\mu}_{\text{forced},1} - \hat{\mu}_{\text{sel}}}{1-p}.$$

with variances

$$\text{Var}(\hat{\tau}_1) = \text{Var}\left[\frac{\hat{\mu}_{\text{sel}} - \hat{\mu}_{\text{forced},0}}{p}\right] = \frac{1}{p^2}(\text{Var}[\hat{\mu}_{\text{sel}}] + \text{Var}[\hat{\mu}_{\text{forced},0}])$$

and

$$\text{Var}(\hat{\tau}_0) = \text{Var}\left[\frac{\hat{\mu}_{\text{forced},1} - \hat{\mu}_{\text{sel}}}{1-p}\right] = \frac{1}{(1-p)^2}(\text{Var}[\hat{\mu}_{\text{forced},1}] + \text{Var}[\hat{\mu}_{\text{sel}}])$$

where I use the fact that the means are estimated on disjoint sets of observations to ignore covariance.

Under the assumptions above along with homoskedasticity of errors, we know that for a sample of size N , $\text{Var}(\sum_{i=1}^N y_i) = \frac{\sigma^2}{N}$. So, we can estimate the variances of each of the terms in the estimator:

$$\begin{aligned} \text{Var}(\hat{\mu}_{\text{sel}}) &= \frac{\sigma^2}{N_{\text{PICA}}/2} = \frac{2\sigma^2}{N_{\text{PICA}}}, \\ \text{Var}(\hat{\mu}_{\text{forced},0}) &= \frac{\sigma^2}{N_{\text{PICA}}/4} = \frac{4\sigma^2}{N_{\text{PICA}}}, \end{aligned}$$

and

$$\text{Var}(\hat{\mu}_{\text{forced},1}) = \frac{4\sigma^2}{N_{\text{PICA}}}.$$

Treating p as fixed,

$$\text{Var}(\hat{\tau}_1) = \frac{1}{p^2} (\text{Var}(\hat{\mu}_{\text{sel}}) + \text{Var}(\hat{\mu}_{\text{forced},0})) = \frac{1}{p^2} \left(\frac{2\sigma^2}{N_{\text{PICA}}} + \frac{4\sigma^2}{N_{\text{PICA}}} \right) = \frac{6\sigma^2}{p^2 N_{\text{PICA}}},$$

so

$$\text{SE}(\hat{\tau}_1) = \frac{\sqrt{6} \sigma}{p \sqrt{N_{\text{PICA}}}}.$$

Similarly,

$$\text{Var}(\hat{\tau}_0) = \frac{1}{(1-p)^2} (\text{Var}(\hat{\mu}_{\text{forced},1}) + \text{Var}(\hat{\mu}_{\text{sel}})) = \frac{6\sigma^2}{(1-p)^2 N_{\text{PICA}}},$$

so

$$\text{SE}(\hat{\tau}_0) = \frac{\sqrt{6} \sigma}{(1-p) \sqrt{N_{\text{PICA}}}}.$$

To obtain a symmetric benchmark where the two PICA HTEs have equal precision, assume $p = 0.5$. Then,

$$\text{SE}(\hat{\tau}_{\text{PICA}}) = \text{SE}(\hat{\tau}_1) = \text{SE}(\hat{\tau}_0) = \frac{\sqrt{6} \sigma}{(1/2) \sqrt{N_{\text{PICA}}}} = 2\sigma \sqrt{\frac{6}{N_{\text{PICA}}}}.$$

B.4 Comparing Efficiency of PICA and CHESSE Estimators

Suppose we have N_{PICA} observations for a PICA design and we select K groups for CHESSE. I solve for the number of observations N_{CHESSE}^* required to estimate treatment effects with equal precision to PICA:

$$\begin{aligned} \text{SE}(\hat{\tau}_{\text{PICA}}) &= \text{SE}(\hat{\tau}_{\text{CHESSE}}) \\ 2\sigma \sqrt{\frac{6}{N_{\text{PICA}}}} &= 2\sigma \sqrt{\frac{K}{N_{\text{CHESSE}}}} \\ \frac{N_{\text{CHESSE}}}{N_{\text{PICA}}} &= \frac{K}{6} \\ N_{\text{CHESSE}}^* &= \frac{K}{6} N_{\text{PICA}} \end{aligned}$$

So, we need $\frac{K}{6}$ times as many observations in order for each CHESSE heterogeneous treatment effect to achieve the same level of precision as each of the two PICA heterogeneous treatment effects.

Note that this is a back-of-the-envelope discussion of estimator precision. In particular, I am ignoring any additional variance created by measuring a proxy for self-selection rather

observing the true self-selection propensity. I am also ignoring the loss of sample in CHES due to participants having high minimum payment requirements.

C Appendix: Reversal of PICA Heterogeneous Treatment Effects

In this appendix, I argue that the heterogeneous treatment effects estimated by PICA and related choice designs may be sensitive to the choice of control condition in such a way that it could reverse the qualitative conclusions drawn from the analysis. In choice designs, researchers estimate heterogeneous treatment effects for groups of participants who, in the selective-exposure condition, opt into and out of treatment. One obvious lever of control that researchers have is the desirability of the control condition. For example, in an experiment measuring the persuasiveness of a Fox News article, the control condition could be unpleasant (e.g., do nothing for a long, fixed period of time), or it could be fun (e.g., read an Entertainment article). Even ignoring that different control conditions could induce different groups of participants to opt out of treatment, it is clear that researchers can manipulate the share of participants who opt out of treatment. This degree of control is problematic because, for some non-linear heterogeneous treatment effect curves, the qualitative conclusions that researchers draw can reverse, since both the sign and magnitude of the estimated heterogeneous treatment effect curves are sensitive to the fraction of participants who opt out of treatment.

[Figure 8](#) illustrates this problem with a hypothetical example. In the first panel, I graph a hypothetical heterogeneous treatment effect curve which is non-linear and largest for participants with a moderate self-selection propensity. The second and third panels of [Figure 8](#) simulate the heterogeneous treatment effects estimated by PICA when the researcher-supplied control is relatively more or less desirable. In panel two, the control condition is more desirable, so those with moderate self-selection propensity opt out, leading the researcher to conclude that the treatment is less effective for those who are likely to select into treatment. In panel three, the control condition is less desirable, so those with moderate self-selection propensity opt in, leading to the opposite conclusion. In this example, changing the desirability of the control condition reverses the conclusion of the PICA analysis.

While the hypothetical example described in [Figure 8](#) is constructed to be a worst-case scenario for choice-based designs, such non-linear treatment effects may arise in natural political communication contexts. For example, consider a piece of right-leaning content. Suppose Republicans are most willing to watch it, moderates somewhat willing, and Democrats unwilling. Additionally, suppose that the content is ineffective for Republicans (who already hold these views) and Democrats (who distrust the source), but persuasive for moderates (who are open to changing their views and do not reflexively distrust the source). This set

of circumstances generates exactly the kind of non-linear treatment effects curve described in [Figure 8](#).

Sensitivity to the control condition is mechanically avoided by CHES, because CHES estimates self-selection propensity into treatment for each participant directly, rather than via choice between treatment and control. Moreover, by eliciting a continuous proxy for self-selection propensity, CHES enables the estimation and detection of non-linear heterogeneous treatment effect curves directly.